

Syllabus Supplement for Students

Organizational Details			
Course Code and Title:	PHYS 708 Theoretical Astroparticle Physics		
Semester:	Fall 2025 (25/08/2025 - 18/12/2025)		
Class Schedule:	Monday, Wednesday: 15:00 - 16:15 Lectures		
Classroom:	Lecture: SAN campus, Room 3016		
Instructors:	Dr. Lecture: Dr. Satyendra Thoudam		
Contact Emails:	satyendra.thoudam@ku.ac.ae		
Office Room Number:	SAN campus: 3010A	Office Telephone Number:	02 312 4218
Office Hours:	Tuesday, Thursday: 13:00 - 14:00 Or send an email for an appointment.		

Assessment Methodology			
		Tentative Dates	Weight
Coursework:	Assignments	See page 3-4	10%
	Quiz 1	See page 3-4	10%
	Quiz 2	See page 3-4	10%
Projects (if applicable)	One project with approval	See page 3-4	20%
Semester Examination(s)	Midterm test	See page-4	20%
Final Examination	Final test	TBA (registrar office)	30%

Instructor Policy on late submission of assignments:

- **Quizzes:** If you miss a quiz, it is your responsibility to contact the Registrar's Office and provide them with your excuse (medical note, etc.). You need to arrange an email to be sent to me (from registrar office). ONLY then your quiz grade will be equal to the average of your other quizzes.
- **Midterm:** There will be no make-up midterm. If you missed this test and you have a valid excuse, the final exam will carry all the point weight of the combined midterm and the final exam (e.g. 50%).

Honor Code Academic Pledge:

"As a Khalifa University student, I will not lie, cheat, steal, or use any unfair means in academic work and will behave according to university rules and UAE societal norms and expectations."

Teaching Plan (Tentative):

Week 1 Aug 25 - Aug 29	Lecture 1: Introduction to high-energy Astrophysical objects. Lecture 2: Messengers in high-Energy Astrophysics.
Week 2 Sep 1 - Sep 5	Lecture 3: Interaction of high-energy particles I (Hadronic processes)
Week 3 Sep 8 - Sep 12	Lecture 3: Interaction of high-energy particles I (Hadronic processes) contd. Lecture 4: Interaction of high-energy particles II (Leptonic processes) <i>Assignment 1 (due Sep 12)</i>
Week 4 Sep 15 - Sep 19	Lecture 5: Particle acceleration: First and Second-order Fermi acceleration
Week 5 Sep 22 - Sep 26	Lecture 6: Particle acceleration at astrophysical shocks
Week 6 Sep 29 - Oct 3	Lecture 7: High-energy emissions (Gamma rays, neutrinos and secondary electrons/positrons). <i>Assignment 2 (due Oct 3)</i>
Week 7 Oct 6 - Oct 10	Lecture 7: High-energy emissions (Gamma rays, neutrinos and secondary electrons/positrons) cont.
Week 8 Oct 13 - Oct 17	<i>Mid-term exam week (tentative)</i>
Week 9 Oct 20 - Oct 24	Lecture 8: Propagation of high-energy particles I (Cosmic-ray protons and nuclei) <i>Assignment 3 (due Oct 24)</i>
Week 10 Oct 27 - Oct 31	Lecture 9: Propagation of high-energy particles II: Electrons. <i>Assignment 4 (due Oct 31)</i>
Week 11 Nov 3 - Nov 7	Lecture 10: Some open problems in cosmic-ray physics.
Week 12 Nov 10 - Nov 14	Lecture 11: Experiments in Astroparticle Physics: Space-based experiments. <i>Assignment 5 (due Nov 14)</i>
Week 13 Nov 17 - Nov 21	Lecture 12: Experiments in Astroparticle Physics: Ground-based experiments
Week 14 Nov 24 - Nov 28	Project week (tentative).
Week 15 Dec 1 - Dec 5	Project week (tentative). Oral presentation of project results.
Dec 6 - Dec 12	<i>Final Exam period</i>

Grading Scheme:

The official Khalifa University of Science and Technology grading system uses letter grades with pluses and minuses. Passing grades range from A to D; F is failing. Each letter grade is assigned a grade point on a four-point scale as set out below:

Letter Grade	Grade Point	Description
A	4.00	Excellent
A-	3.70	Very Good
B+	3.30	
B	3.00	Good
B-	2.70	
C+	2.30	Satisfactory
C	2.00	
C-	1.70	Less Than Satisfactory
D	1.00	Poor
F	0.00	Fail

The following is indicative of typical ranges associated with letter grades for the College of Arts & Sciences undergraduate courses, and is the default setting in the Gradebook:

A	4.00	92.5%-100%	Excellent
A-	3.70	89.5% to < 92.5%	Very Good
B+	3.30	86.5% to < 89.5%	
B	3.00	82.5% to < 86.5%	Good
B-	2.70	79.5% < 82.5%	
C+	2.30	76.5% to < 79.5%	Satisfactory
C	2.00	72.5% to < 76.5%	
C-	1.70	69.5% to < 72.5%	Less Than Satisfactory
D	1.00	59.5% to < 69.5%	Poor
F	0.00	< 59.5%	Fail